Auteur: Yoren Collier
Studierichting: Informatica
Oefeningenreeks 4A nr. 37.14

$$I = \int \frac{4dx}{x^2 + x + 1} dx = \int \frac{4dx}{x^2 + x\frac{1}{4} + \frac{3}{4}} dx = \int \frac{4dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx$$

Stel: $t = x + \frac{1}{2} \Rightarrow dt = dx$

$$= 4 \int \frac{1}{t^2 + \sqrt{\frac{3}{4}}} dt = 4 \cdot \frac{1}{\sqrt{\frac{3}{4}}} \operatorname{Bgtan}\left(\frac{t}{\sqrt{\frac{3}{4}}}\right) + c$$

$$= \frac{8}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + c = \frac{8\sqrt{3}}{3} \operatorname{Bgtan}\left(\frac{(2x+1)\sqrt{3}}{3}\right) + c$$

References